

# Reviewer Pack — Finite Geometry Rank and $ACC^0$ Circuit Size for 3-SAT Instan...

Entry 8255cc53c91e · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-01 16:47:22 UTC

## Finite Geometry Rank and $ACC^0$ Circuit Size for 3-SAT Instances

**Entry ID:** 8255cc53c91e **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-01 16:47:22 UTC

### 1. Conjecture

**Field A** (mathematical branch): Finite Geometry **Field B** (complexity object): Boolean Circuit Size ( $ACC^0$ )

#### Statement:

For any 3-SAT instance  $\Phi$  with  $n$  variables, the rank of its incidence matrix over  $GF(2)$  is  $\Theta(\log n)$  if and only if  $\Phi$  can be computed by an  $ACC^0$  circuit of size  $O(n^\epsilon)$  for some  $\epsilon < 1$ .

#### Rationale (proposer's reasoning):

The incidence matrix's rank over  $GF(2)$  captures linear dependencies among clauses, which may constrain  $ACC^0$  circuits' ability to simulate  $\Phi$ . High rank implies complex dependencies, requiring larger circuits, while low rank suggests structured dependencies amenable to small  $ACC^0$  implementations.

**Taxonomy category:** ACC\_LB\_via\_WILLIAMS (status at proposal time: partially\_alive)

### 2. Pre-registration (Popper-style)

**Hash (SHA-256 prefix):** b881c5c32bfff70a

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

#### Acceptance criterion:

default:  $\geq 80\%$  seeds must support, no counterexample

### 3. Barrier filter (F1)

**Final:** PASS

| Barrier        | Verdict | Confidence | LLM <sub>1</sub> | LLM <sub>2</sub> |
|----------------|---------|------------|------------------|------------------|
| RELATIVIZATION | SAFE    | 0.00       | UNCERTAIN        | UNCERTAIN        |
| NATURAL_PROOFS | SAFE    | 0.00       | UNCERTAIN        | UNCERTAIN        |

| Barrier       | Verdict | Confidence | LLM <sub>1</sub> | LLM <sub>2</sub> |
|---------------|---------|------------|------------------|------------------|
| ALGEBRIZATION | SAFE    | 0.00       | UNCERTAIN        | UNCERTAIN        |
| KARP_LIPTON   | SAFE    | 0.00       | UNCERTAIN        | UNCERTAIN        |

## 4. Novelty audit

**Verdict:** NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

## 5. Empirical test harness

**Multi-seed protocol:** 5 seeds (11, 23, 37, 53, 71)

**Sandbox:** isolated subprocess, stdlib-only,  $\leq 90$ s wall time per cycle **Execution:** rc=1, elapsed=0.1s

### 5.1 Generated Python source

```
import random
import math

def gaussian_elimination(matrix):
    n = len(matrix)
    for i in range(n):
        # Find pivot
        max_row = i
        for j in range(i+1, n):
            if abs(matrix[j][i]) > abs(matrix[max_row][i]):
                max_row = j
        matrix[i], matrix[max_row] = matrix[max_row], matrix[i]

        # Eliminate below the pivot
        for j in range(i+1, n):
            factor = matrix[j][i] / matrix[i][i]
            for k in range(n):
                matrix[j][k] -= factor * matrix[i][k]

    rank = 0
    for row in matrix:
        if any(row):
            rank += 1
    return rank

def generate_3sat_instance(n, m):
    variables = list(range(1, n+1))
    clauses = []
    for _ in range(m):
        clause = random.sample(variables, 3)
        clauses.append(clause)
    return clauses

def run_trial(seed: int) -> dict:
    random.seed(seed)

    results = []
    for n in [5, 10, 15, 20, 30, 40]:
        instances_tested = 0
```

```

total_rank = 0

for _ in range(5): # Sample 5 instances per n
    clauses = generate_3sat_instance(n, 3*n)
    incidence_matrix = [[int(v in clause) for v in range(1, n+1)] for clause in clauses]

    rank = gaussian_elimination(incidence_matrix)
    total_rank += rank
    instances_tested += 1

avg_rank = total_rank / instances_tested
conjecture_holds = False
counterexample = ""

if abs(avg_rank - math.log(n, 2)) < 0.5:
    conjecture_holds = True

return {
    "metric_name": "rank",
    "metric_value": avg_rank,
    "instances_tested": instances_tested * len([5, 10, 15, 20, 30, 40]),
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [int(math.log(p)) for p in [2, 3, 5, 7, 11, 13, 17, 19]]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_rank = sum(result["metric_value"] for result in results) / len(results)
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_rank} std=unknown support_fraction={support_fraction}")
    elif any(not result["conjecture_holds"] for result in results):
        counterexample = next(result for result in results if not result["conjecture_holds"])["counterexample"]
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE mapping_undefined")

```

## 6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

## 7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ba09c122.py", line 85, in <module>
    result = run_trial(seed)
              ^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ba09c122.py", line 60, in run_trial
    rank = gaussian_elimination(incidence_matrix)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_ba09c122.py", line 32, in gaussian_elim
    matrix[j][k] -= factor * matrix[i][k]
               ^^^^^^^^^^^^^
~~~~~^^^

```

IndexError: list index out of range

**8. Critic adversarial review**

**Critic verdict:** CHALLENGE

**Critic reasoning:**  
 skipped: test crashed before producing data

**9. Final verdict & safety rail**

**Verdict:** INCONCLUSIVE

**Reasoning:**  
 Test crashed during Gaussian elimination, preventing data collection. Support condition cannot be evaluated without successful runs. | next: Fix the Gaussian elimination implementation to handle matrix dimensions properly and re-run tests

**11. Audit log (LLM calls)**

**Total LLM calls:** 7

| # | Phase           | Provider      | Model            | Tokens in | out | Latency (ms) |
|---|-----------------|---------------|------------------|-----------|-----|--------------|
| 1 | propose         | ollama_remote | qwen3:8b         | 0         | 0   | 82604        |
| 2 | preregistration | ollama_remote | qwen3:8b         | 0         | 0   | 24091        |
| 3 | novelty         | ollama_remote | qwen3:8b         | 0         | 0   | 20630        |
| 4 | novelty         | ollama_remote | qwen3:8b         | 0         | 0   | 15146        |
| 5 | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 13467        |
| 6 | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 9523         |
| 7 | judge           | ollama_remote | qwen3:8b         | 0         | 0   | 16208        |

**Totals:** 0 input tokens, 0 output tokens, ~\$0.0000 cost, 181670 ms total latency. Provider mix: {'ollama\_remote': 7}  
*(full prompt+response transcripts available in research/audit/8255cc53c91e.jsonl)*

**12. Reproducibility**

- To reproduce this cycle:
1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
  2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
  3. The full LLM transcripts are in research/audit/8255cc53c91e.jsonl for verification of provenance
  4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

**Sandbox file:** research/pvsnp\_sandbox/test\_\*.py (per-cycle)

**Replay tarball:** research/replay/8255cc53c91e.tar.gz (if generated)